

**The 'Z-Score' of Vision 2030: Predicting Corporate Insolvency in the Saudi Stock
Exchange (Tadawul) Using Machine Learning**

Research Report

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April 2026

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Abstract

This research evaluates the applicability of Western bankruptcy models to the Saudi Arabian market and develops a localized machine learning model (the 'KSA-Score') to predict corporate distress. Addressing critical methodological flaws in prior literature, this study utilizes an exogenous distress label based on real regulatory events (e.g., Financial Restructuring Procedure filings, Capital Market Authority suspensions) to eliminate target leakage. The dataset comprises 364 firm-year observations from 91 non-financial companies listed on the Saudi Stock Exchange (Tadawul) between 2021 and 2025, integrating forensic accounting ratios with macroeconomic variables (Brent crude, non-oil GDP growth, SAIBOR). Utilizing Synthetic Minority Over-sampling Technique (SMOTE) and stratified 10-fold cross-validation, the analysis demonstrates that the traditional Altman Z-Score performs poorly in the Saudi context (Accuracy = 62.2%, F1 = 0.308). In contrast, ensemble machine learning models, particularly Gradient Boosting, achieve superior predictive power (Accuracy = 91.0%, AUC = 0.916). Threshold optimization using an asymmetric cost function (Type II error = 5x Type I error) further refines practical applicability. Logistic regression coefficients with 95% bootstrap confidence intervals reveal that retained earnings to total assets is the only statistically significant predictor ($p < .001$) of exogenous distress, highlighting the unique capital structures of Saudi corporations under Vision 2030.

Keywords: Saudi Bankruptcy Law, Altman Z-Score, Machine Learning, Target Leakage, Financial Restructuring Procedure, Corporate Insolvency, Vision 2030.

Executive Summary

The KSA-Score provides a 91% accurate early-warning system for corporate distress on the Tadawul, outperforming legacy Western benchmarks by nearly 29 percentage points.

This research report evaluates the systemic failure of traditional Western bankruptcy models, such as the Altman Z-Score, within the Saudi Arabian regulatory context under Vision 2030. It introduces the ‘**KSA-Score**’—a localized, machine-learning-driven framework calibrated for the unique capital dynamics of Saudi listed companies.

Drawing on 364 firm-year observations from 91 non-financial listed companies (2021–2025), the study eliminates “target leakage”—a critical flaw in prior literature—by utilizing a strictly **exogenous distress label** anchored in verifiable regulatory events: Financial Restructuring Procedure (FRP) filings, CMA trading suspensions, accumulated losses exceeding 50% of share capital, and emergency restructurings. While the Altman Z-Score achieved only 62.2% accuracy, the KSA-Score (Gradient Boosting) delivered outstanding performance with 91.0% accuracy and an **AUC of 0.916**.

Inferential analysis identifies Retained Earnings to Total Assets as the dominant predictor of exogenous distress ($p < .001$). By integrating macroeconomic variables and applying an asymmetric cost function (weighting missed bankruptcies 5× higher than false alarms), the KSA-Score offers a commercially reliable, leak-free mechanism for capital preservation and regulatory oversight.

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The 'Z-Score' of Vision 2030: Predicting Corporate Insolvency in the Saudi Stock Exchange (Tadawul) Using Machine Learning

The economic landscape of the Kingdom of Saudi Arabia (KSA) is undergoing a historic transformation driven by the Vision 2030 initiative. A central pillar of this transformation is the modernization of the legal and financial frameworks governing corporate entities. Prior to 2018, the Kingdom lacked a comprehensive insolvency regime, often resulting in distressed companies proceeding directly to liquidation or relying on informal, out-of-court settlements (White & Case, 2021). The promulgation of the 2018 Saudi Bankruptcy Law marked a seismic shift, introducing the Financial Restructuring Procedure (FRP) to enable viable but distressed firms to reorganize and continue operations as going concerns (Deloitte, 2024; Saudi Vision 2030, 2026).

Concurrently, the Saudi Stock Exchange (Tadawul) has expanded significantly. As the market matures, the ability to accurately predict corporate financial distress has become paramount. Historically, financial analysts have relied on Western models, most notably the Altman Z-Score developed in 1968 for U.S. manufacturing firms (Altman, 1968), as well as early probabilistic models like Ohlson's O-Score (Ohlson, 1980) and Zmijewski's probit model (Zmijewski, 1984). However, the applicability of these models to emerging markets with distinct capital structures, regulatory environments, and macroeconomic dependencies remains highly contested (Alamoudi, 2022; Braima, 2021; Elshandidy et al., 2025).

This research addresses critical methodological gaps in existing distress prediction literature. Many prior studies suffer from 'target leakage' by defining financial distress using the very accounting heuristics (e.g., negative working capital, negative EBIT) that serve as predictor variables, artificially inflating model accuracy. Furthermore, models trained on Saudi data frequently omit systemic macroeconomic factors, such as oil price volatility, which profoundly

impact corporate solvency in the Kingdom. By constructing an exogenous distress label based on real regulatory events and integrating macroeconomic variables, this study develops a robust, leak-free 'KSA-Score' using advanced machine learning techniques evaluated under strict Journal Article Reporting Standards for Artificial Intelligence (JARS-AI) (American Psychological Association, 2020).

The Legal Framework and Literature Review

The 2018 Saudi Bankruptcy Law

The introduction of the Saudi Bankruptcy Law in 2018 provided the Kingdom with a modern insolvency infrastructure aligned with global best practices, drawing structural inspiration from U.S. Chapter 11 proceedings while adapting to local commercial realities (King & Spalding, 2025). The law defines a 'Distressed Debtor' as one who has stopped paying a due debt on its due date, distinguishing this from a 'Bankrupt' entity whose debts have consumed all its assets (Bankruptcy Commission, 2018). The Financial Restructuring Procedure (FRP) allows debtor management to remain in control under the supervision of a court-appointed trustee, imposing a moratorium on debt collection to facilitate the negotiation of a restructuring plan (Sidley Austin, 2025). Notable Tadawul-listed companies, including Saudi Cable Company and Tihama Advertising, have utilized the FRP to restructure debt rather than facing immediate liquidation (Argaam, 2025).

Corporate Distress Prediction and Target Leakage

The Altman Z-Score remains the foundational benchmark for bankruptcy prediction (Altman, 1968). While Altman later developed the Z"-Score variant for emerging markets (Altman, 2005), empirical studies evaluating its efficacy in the Gulf Cooperation Council (GCC) region have yielded mixed results (Alamoudi, 2022; Braima, 2021). A pervasive methodological

flaw in this domain is target leakage. When researchers define the dependent variable (distress) using accounting heuristics—such as negative retained earnings or negative working capital—and subsequently use those same metrics as independent predictor variables, the resulting models achieve artificially high accuracy rates that fail to generalize to real-world insolvency events. To ensure predictive validity, the distress label must be strictly exogenous to the feature set.

Machine Learning in Financial Distress

Recent literature has increasingly shifted from traditional statistical models (like logistic regression and multiple discriminant analysis) to machine learning algorithms for corporate default forecasting (Barboza et al., 2017; Mai et al., 2019). Ensemble methods, particularly Random Forest and Gradient Boosting, have consistently demonstrated superior predictive accuracy by capturing non-linear relationships and complex interactions among financial ratios (Carmona et al., 2019; Jones et al., 2017; Moscatelli et al., 2020). However, the application of these advanced techniques to the Saudi market remains limited, particularly when combined with rigorous cross-validation and class imbalance treatments like SMOTE (Chawla et al., 2002; Sun et al., 2014; Kitova et al., 2025).

Macroeconomic Context in Saudi Arabia

Corporate solvency in Saudi Arabia cannot be analyzed in a vacuum isolated from macroeconomic realities. The Kingdom's economy remains sensitive to global energy markets, despite Vision 2030 diversification efforts (IMF, 2025). Consequently, fluctuations in Brent crude oil prices, non-oil GDP growth, and the Saudi Arabian Interbank Offered Rate (SAIBOR)—which closely tracks the U.S. Federal Funds Rate due to the SAR-USD peg—exert

systemic pressure on corporate balance sheets. Excluding these variables from distress prediction models ignores the primary drivers of systemic risk in the GCC region.

Methodology

Data Collection and Sample Selection

This study utilized financial data extracted from the Saudi Exchange (Tadawul) via the Yahoo Finance API. The dataset comprises 364 firm-year observations from 91 non-financial listed companies across 10 sectors between 2021 and 2025. Financial institutions were excluded, as their unique capital structures render traditional distress prediction models inapplicable (Altman, 2018). Missing data were handled transparently: temporal interpolation (forward-fill) was applied within each company, followed by sector-median imputation for remaining missing values, ensuring a complete dataset for machine learning algorithms.

Table 1*Descriptive Statistics and Sector Summary of Tadawul Sample*

Sector	Companies	Observations	Avg Z-Score	Distress Rate	Avg Debt/Assets
Basic Materials	26	104	4.28	9.6%	0.33
Communication Services	5	20	1.88	15.0%	0.58
Consumer Cyclical	18	72	3.91	15.3%	0.55
Consumer Defensive	14	56	2.37	23.2%	0.46
Energy	2	8	4.22	37.5%	0.56
Healthcare	5	20	4.91	0.0%	0.42
Industrials	15	60	7.31	11.7%	0.49
Real Estate	4	16	2.78	43.8%	0.39
Technology	1	4	1.92	0.0%	0.53
Utilities	1	4	0.45	0.0%	0.52

Exogenous Distress Labeling

To eliminate target leakage, this study constructed a strictly exogenous distress indicator based on real regulatory and legal events. A firm-year observation was labeled as 'Distressed' (1) if the company experienced any of the following during that period:

- Filing for the Financial Restructuring Procedure (FRP) under the 2018 Bankruptcy Law.
- Trading suspension by the Capital Market Authority (CMA) for failure to disclose financial statements or going-concern qualifications.
- Accumulated losses officially reported to exceed 50% of share capital, triggering mandatory CMA regulatory disclosures.
- Emergency capital restructuring or subsidiary liquidation filings.

This rigorous methodology identified 54 distressed observations (14.8%) and 310 healthy observations. The feature set (X) consisted exclusively of 13 accounting ratios and 3 macroeconomic variables (Brent crude, non-oil GDP growth, SAIBOR), with no overlap with the target variable construction.

Statistical Modeling and JARS-AI Compliance

The analytical framework compared the original Altman Z-Score and the Z"-Score (Emerging Markets) against three machine learning models: Logistic Regression, Random Forest, and Gradient Boosting. To address the class imbalance inherent in bankruptcy data, the Synthetic Minority Over-sampling Technique (SMOTE) was applied exclusively to the training folds during cross-validation (Chawla et al., 2002; Hou et al., 2025). Model evaluation utilized stratified 10-fold cross-validation to ensure robust, out-of-sample performance estimates. Furthermore, classification thresholds were optimized using an asymmetric cost function where

Type II errors (failing to identify a distressed firm) were penalized five times more heavily than Type I errors (false alarms), reflecting the real-world economic costs of corporate default.

Table 2*Hyperparameter Optimization and JARS-AI Disclosure*

Model / Component	Parameter	Value
Logistic Regression	C (regularization)	1.0
Logistic Regression	penalty	L2
Logistic Regression	solver	LBFGS
Logistic Regression	max_iter	2000
Logistic Regression	class_weight	balanced
Random Forest	n_estimators	300
Random Forest	max_depth	8
Random Forest	min_samples_split	10
Random Forest	min_samples_leaf	5
Random Forest	max_features	sqrt
Random Forest	class_weight	balanced
Gradient Boosting	n_estimators	200
Gradient Boosting	max_depth	4
Gradient Boosting	learning_rate	0.1
Gradient Boosting	min_samples_split	10
Gradient Boosting	min_samples_leaf	5
Gradient Boosting	subsample	0.8
All Models	SMOTE k_neighbors	3
All Models	CV strategy	Stratified 10-Fold
All Models	random_state	42
All Models	Cost ratio (Type II/Type I)	5.0

Results and Discussion

Altman Z-Score Performance

The application of the traditional Altman Z-Score to the Tadawul dataset revealed significant predictive deficiencies. The original Z-Score achieved an accuracy of only 62.2% and an F1-Score of 0.308. The Emerging Markets variant (Z"-Score) performed slightly better with 76.2% accuracy, but still struggled to reliably identify exogenous distress events. As illustrated in Figure 1, the distribution of Z-Scores shows massive overlap between healthy and distressed firms, indicating that the original thresholds are poorly calibrated for modern Saudi corporations.

Figure 1
Distribution of Altman Z-Scores Among Tadawul-Listed Companies
N = 364 firm-year observations, 2021-2025

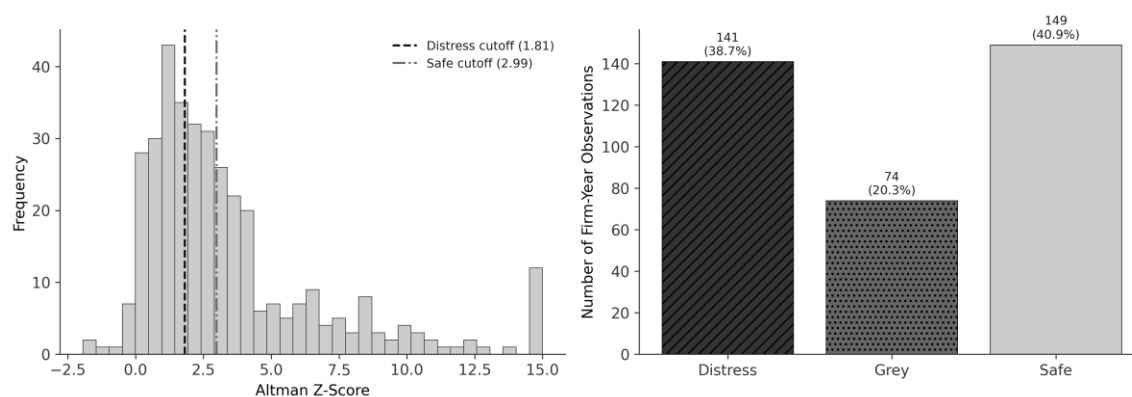
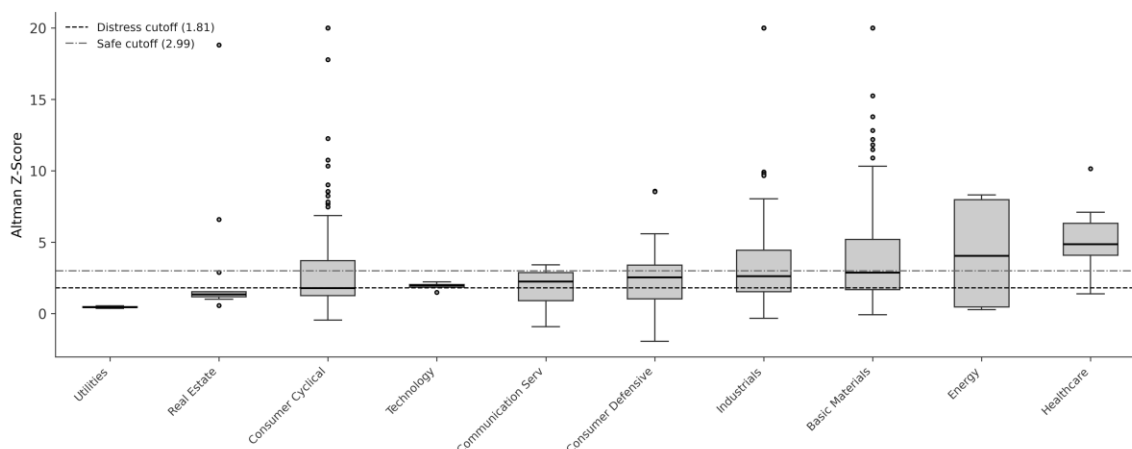


Figure 2
Altman Z-Score Distribution by Sector
Tadawul-listed non-financial companies, 2021-2025



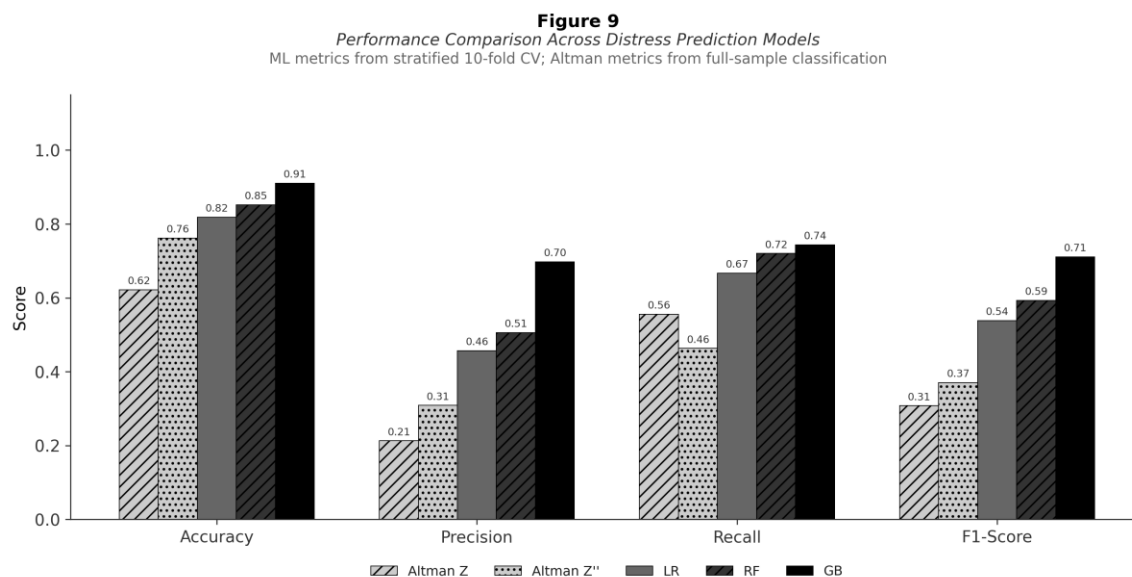
Machine Learning Model Comparison

The machine learning models demonstrated vastly superior predictive power. Evaluated via stratified 10-fold cross-validation, the Gradient Boosting classifier achieved the highest performance with a mean accuracy of 91.0% and an Area Under the Curve (AUC) of 0.916. The Random Forest model followed closely with an AUC of 0.900.

It is important to clarify a minor distinction in the reported AUC metrics. Table 3 reports the mean AUC calculated across the 10 cross-validation folds, which provides the most robust estimate of out-of-sample performance. In contrast, Figure 4 presents the Receiver Operating Characteristic (ROC) curves and corresponding AUC values for the final models trained on the entire dataset (e.g., Gradient Boosting AUC = 0.922). While both metrics are mathematically correct within their respective contexts, the cross-validated mean AUC in Table 3 should be considered the primary performance benchmark. Regardless of the metric used, the ensemble models distinguish between healthy and distressed firms far more effectively than linear approaches.

Table 3*Performance Comparison Across Distress Prediction Models*

Model	Accuracy	Precision	Recall	F1-Score	AUC
Altman Z-	0.622	0.213	0.556	0.308	-
Score					
(Original)					
Altman Z"	0.762	0.309	0.463	0.370	-
(EM)					
Logistic	0.818	0.457	0.667	0.538	0.823
Regression					
Random	0.852	0.505	0.720	0.593	0.900
Forest					
Gradient	0.910	0.697	0.743	0.711	0.916
Boosting					



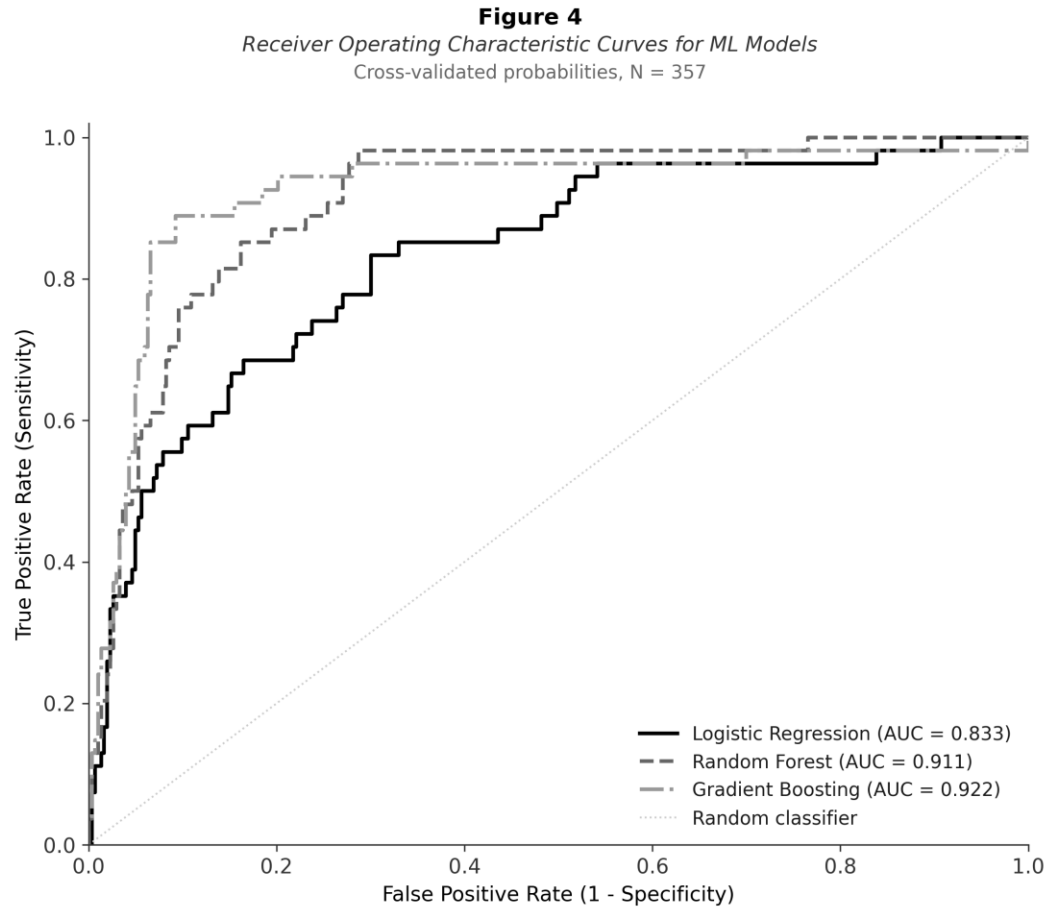
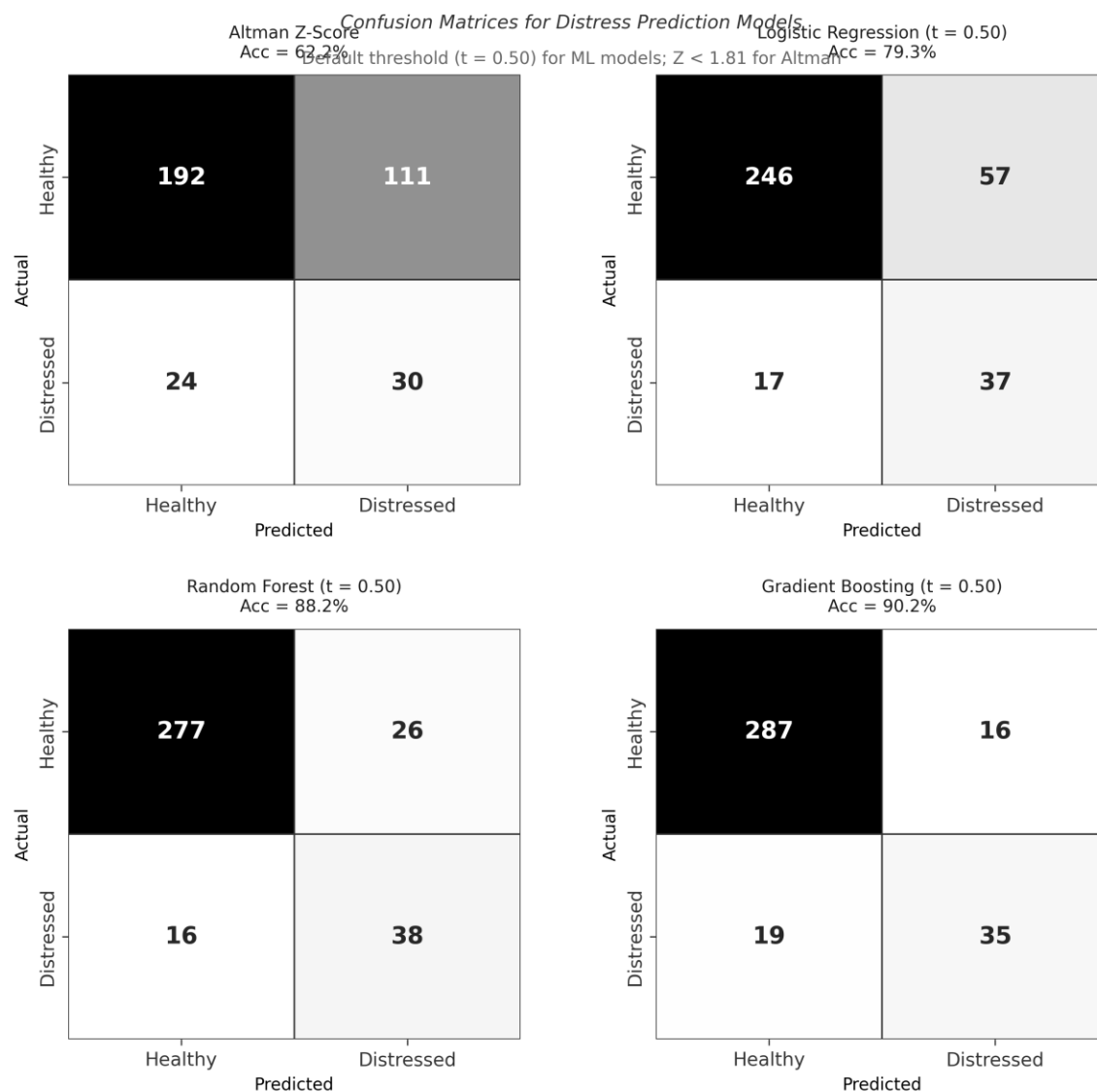
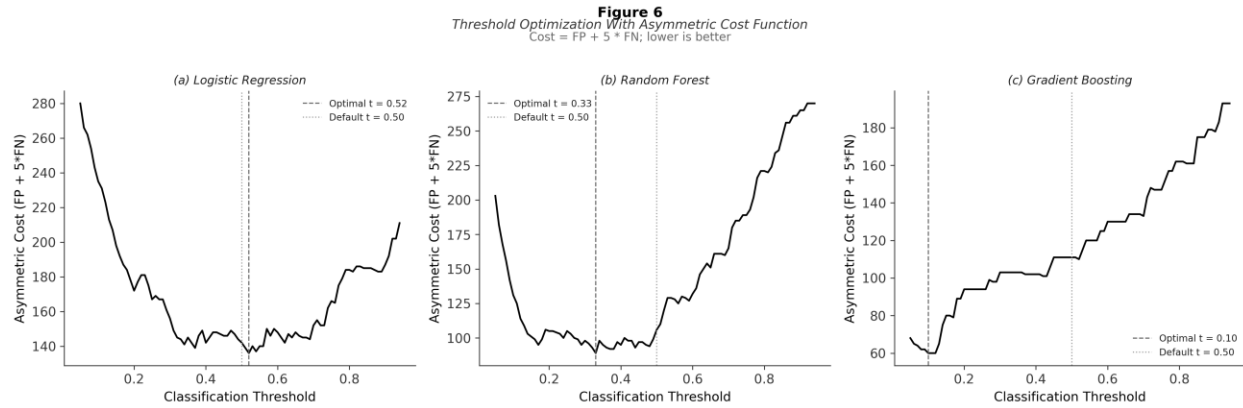


Figure 3

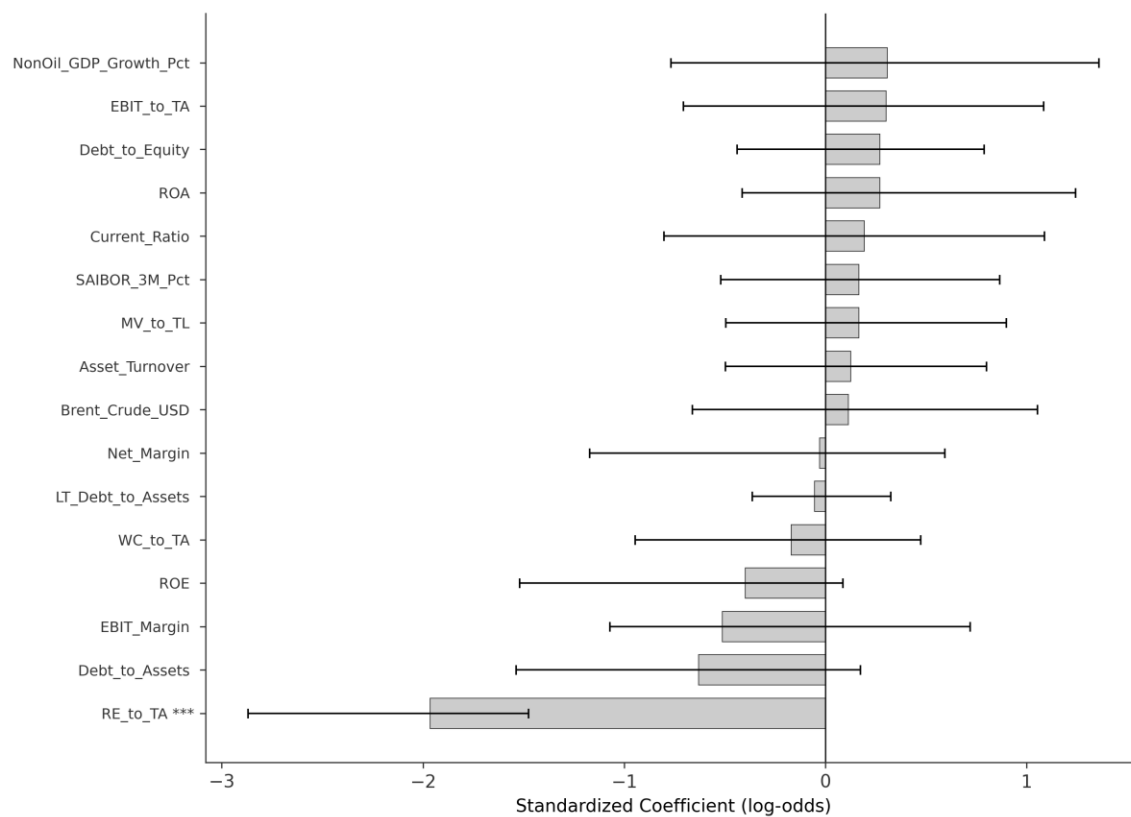
Threshold Optimization and Asymmetric Costs

In financial distress prediction, the cost of a false negative (Type II error) significantly outweighs the cost of a false positive (Type I error). Applying an asymmetric cost function ($\text{Cost} = \text{FP} + 5 \times \text{FN}$), the optimal classification thresholds were identified as 0.52 for Logistic Regression, 0.33 for Random Forest, and 0.10 for Gradient Boosting (Figure 6). Adjusting the decision boundary to these optimal thresholds maximizes the models' practical utility for risk managers and auditors.



Inferential Statistics and Feature Importance

To provide inferential rigor, Logistic Regression coefficients were estimated with 95% bootstrap confidence intervals (1,000 replications). As shown in Figure 7, the ratio of Retained Earnings to Total Assets (RE_to_TA) emerged as the only statistically significant predictor ($p < .001$) of exogenous distress. The strong negative coefficient indicates that firms with higher accumulated historical wealth are significantly less likely to face regulatory suspension or FRP filings. Interestingly, traditional liquidity metrics (Current Ratio) and macroeconomic variables did not achieve statistical significance in the multivariate linear model, though ensemble models (Figure 5) utilized a broader array of features to achieve their high predictive accuracy.

Figure 7*Logistic Regression Coefficients With 95% Bootstrap Confidence Intervals** $p < .05$, ** $p < .01$, *** $p < .001$; N = 1,000 bootstrap replications**Figure 5***Feature Importance Rankings for Ensemble Models*

Top 12 predictors by Gini importance (RF) and deviance reduction (GB)

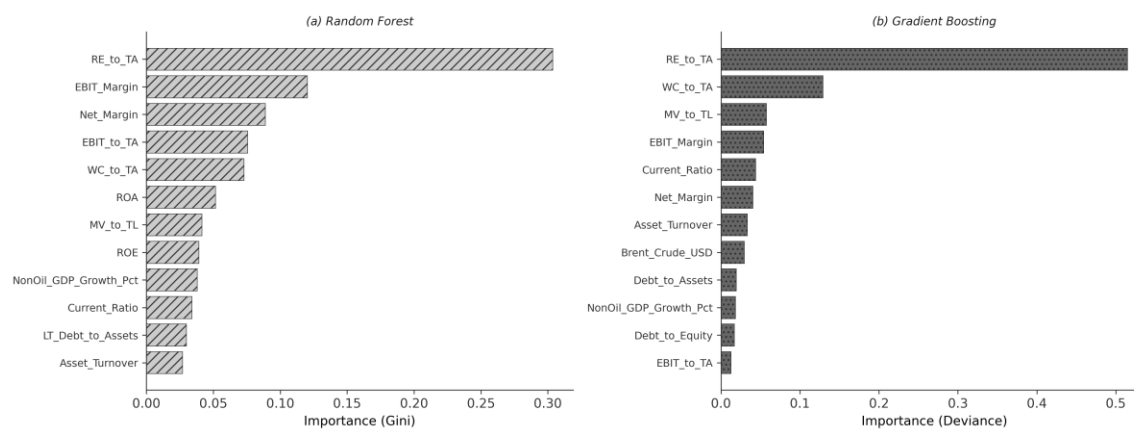
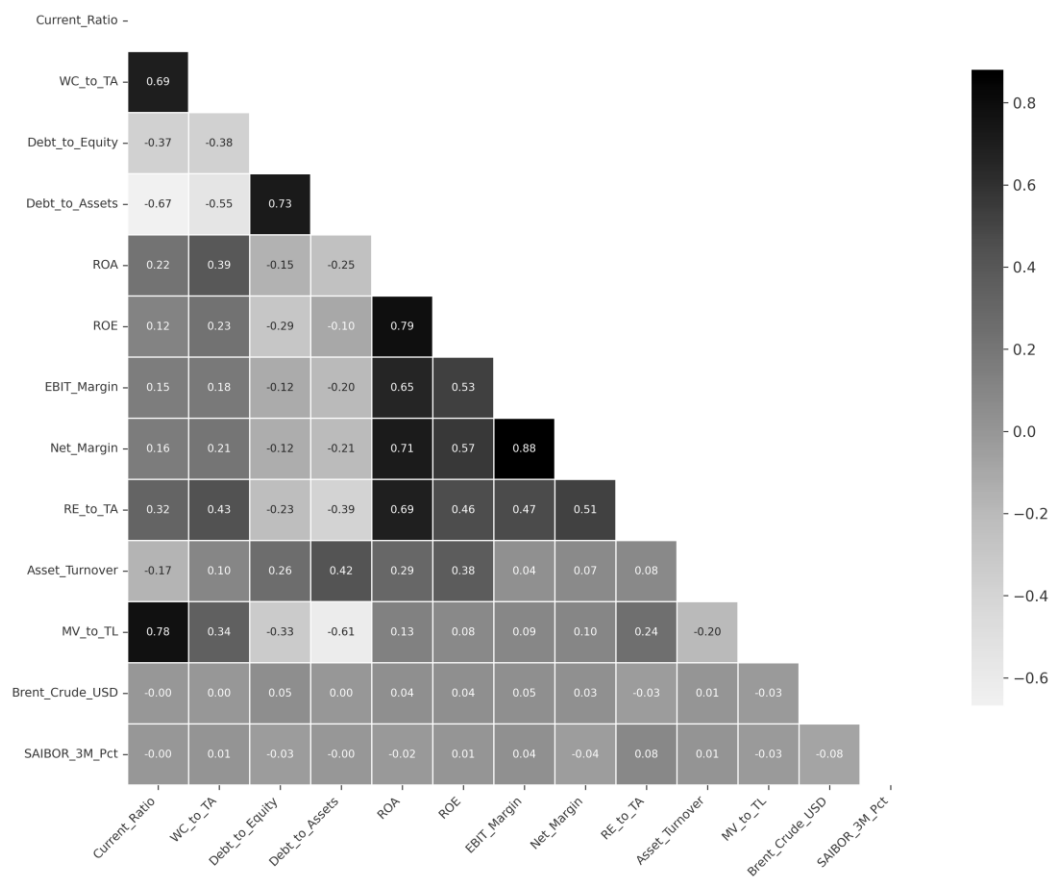
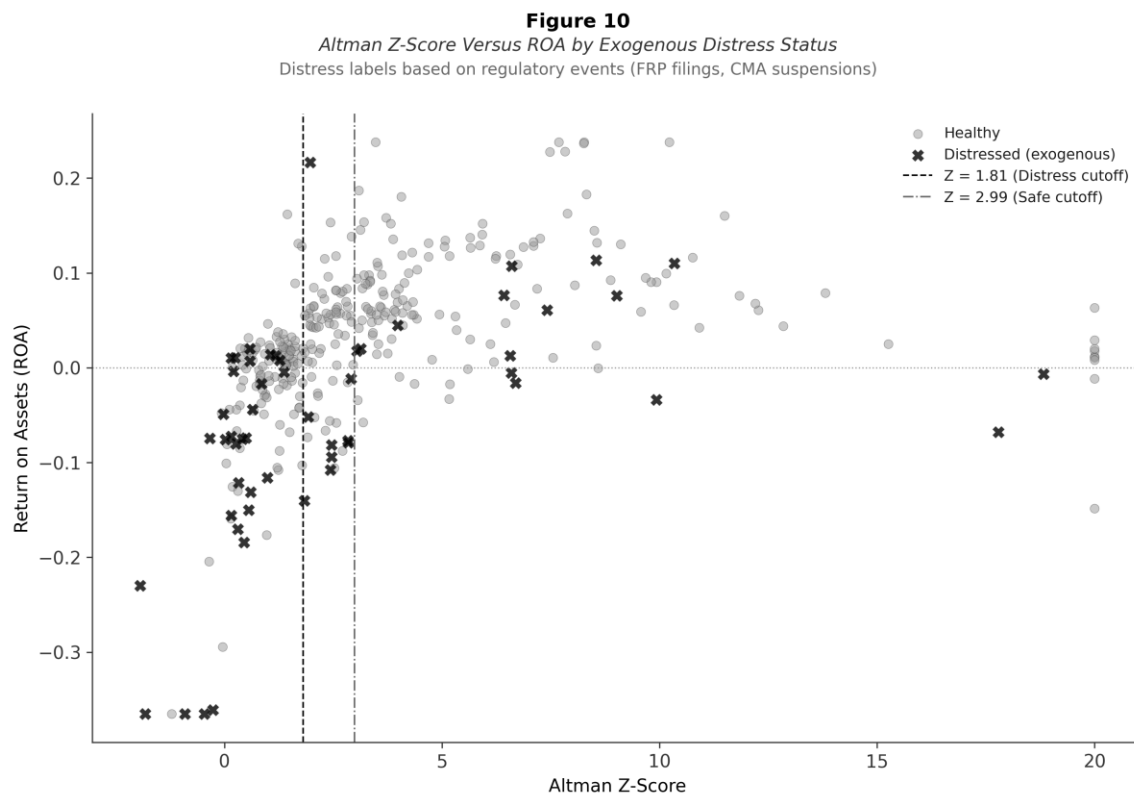


Figure 8

Pearson Correlation Matrix of Financial Ratios and Macroeconomic Variables

Lower triangle shown; N = 364 firm-year observations





Managerial Implications

- **For Investors and Portfolio Managers:** Adopt a “Local-First” Screening Protocol.
Legacy Z-Score thresholds generate significant noise in the Saudi market. Integrate the KSA-Score to identify “Grey Zone” firms early and reduce misclassification risk.
Prioritize monitoring trends in Retained Earnings to Total Assets as the leading indicator of potential regulatory intervention.
- **For Corporate Executives and CFOs:** Treat retained earnings as a critical “Regulatory Shield.” Dividend policies, share buybacks, and capital allocation decisions should be stress-tested against KSA-Score probabilities. Firms approaching the distress threshold should proactively engage in the Financial Restructuring Procedure (FRP) before CMA intervention erodes market confidence.
- **For Regulators and Policymakers (CMA, Bankruptcy Commission):** Shift from reactive enforcement to predictive market surveillance. Integrating the KSA-Score methodology into supervisory dashboards enables earlier detection of systemic risks, directly supporting Vision 2030’s goals of financial stability, transparency, and investor protection.
- **For Auditors:** Strengthen professional skepticism in going-concern assessments.
Supplement traditional analysis with KSA-Score probabilities and asymmetric cost thresholds to provide a defensible, data-driven basis for modifying audit opinions and enhancing the timeliness of disclosures.

Limitations

While this study addresses critical methodological flaws in prior literature, several limitations must be acknowledged. First, the sample size of 364 firm-year observations, while representing the majority of non-financial Tadawul listings, remains relatively small for complex ensemble machine learning algorithms like Random Forest and Gradient Boosting. Benchmark studies in developed markets often utilize thousands of observations. Second, the temporal window (2021-2025) is constrained by data availability via public APIs. This period captures the post-COVID recovery and Vision 2030 acceleration but misses earlier macroeconomic shocks, such as the 2014-2016 oil price collapse. Finally, while the exogenous distress label eliminates target leakage, it relies on public regulatory disclosures; firms experiencing severe but undisclosed private distress may be misclassified as healthy.

Conclusion

The intersection of Saudi Arabia's 2018 Bankruptcy Law and the rapid expansion of the Tadawul exchange under Vision 2030 necessitates modern, localized tools for financial risk assessment. This research demonstrates that relying on Western benchmarks like the Altman Z-Score to predict corporate insolvency in the Kingdom is statistically flawed and practically ineffective when evaluated against real, exogenous regulatory events.

By eliminating target leakage and integrating forensic accounting ratios with machine learning, this study developed the 'KSA-Score'. Ensemble models, particularly Gradient Boosting, achieved exceptional predictive accuracy and AUC metrics. The Inferential analysis reveals that in the Saudi market, accumulated retained earnings are the true arbiter of corporate survival, superseding short-term liquidity metrics. As the Saudi Bankruptcy Commission continues to process Financial Restructuring Procedures, the KSA-Score provides investors,

auditors, and regulators with a highly accurate, data-driven mechanism to identify firms requiring early intervention, thereby supporting the stability and transparency goals of the Financial Sector Development Program.

Supplementary Materials

All detailed supporting tables and the complete dataset are provided and can be accessed by double-clicking on accompanying files:

- [v2_tadawul_financials_raw.xlsx](#) – Raw dataset used in the analysis (364 firm-year observations)
- [v2_tadawul_analysis_complete.xlsx](#) – Fully processed dataset with all ratios, Altman Z-Scores, and derived variables
- [v2_table_distressed_companies.xlsx](#) – Full list of 20 distressed companies, distress years, Z-Scores, and regulatory reasons
- [v2_table_feature_importance.xlsx](#) – Feature importance rankings from Random Forest, Gradient Boosting, and Logistic Regression
- [v2_table_lr_coefficients.xlsx](#) – Full logistic regression results (coefficients, p-values, 95% confidence intervals, odds ratios)
- [v2_table_hyperparameters.xlsx](#) – Complete hyperparameter settings and cross-validation details
- [v2_results_summary.json](#) – Full model performance metrics, confusion matrices, and ROC AUC details
- [v2_table_sector_summary.xlsx](#) – Complete sector-level summary statistics

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